Series

Chapter Overview

Solution 2

A series is a sum of a finite or infinite number of terms. In Pure Year 1, Σ ("capital sigma") denotes summation.

What we will cover:

1. First n integers: $\sum_{r=1}^{n} r$.
2. First n squares: $\sum_{r=1}^{n} r^2$.
3. First n cubes: $\sum_{r=1}^{n} r^3$.
4. Breaking down complex sums: $\sum_{r=1}^{n}(3 - 2r^2 + 4r)$.
5. Other bounds: $\sum_{r=4}^{2n} r^2$.

We will later prove some results using proof by induction.

From History of Mathematical Notations, Vol. 2

438. The sign Σ for summation is due to L. Euler[1] (1755), who says, "summam indicabimus signo Σ." This symbol was used by Lagrange,[2] but otherwise received little attention during the eighteenth century. A widely used notation for the sum of a series is the capital letter S; it is given, for instance, in G. S. Klügel's Wörterbuch.[3] The Σ to express "sum" occurs in 1829 in Fourier's Theory of Heat,[4] published in 1822, and in C. G. J. Jacobi's[5] elliptic functions of 1829.

Cauchy[6] used three indices m, n, r, as in $\sum_{m}^{n} fr$. Alfred Pringsheim[7] marks the sum of an infinite series thus, $\sum_{0}^{\infty} a_r$.

Jahnke[8] adopts in one place, as a substitute for Σa_ν, the simpler sign $\bar{a}$, which he borrows from geodesists, to designate the sum $a_1 + a_2 + a_3 + \cdots$. Additional symbols for sum are given in §410.

Recap

Solution 3

Determine the results by explicitly writing out the terms in each sum.

$$\sum_{p=3}^{8} p^2$$

$= 3^2 + 4^2 + \cdots + 8^2 = 199.$

$$\sum_{r=0}^{5} (7r+1)^2$$

$= 1^2 + 8^2 + \cdots + 36^2 = 2911.$

Sum of Ones, Integers, Squares and Cubes

Solution 4

These are the four essential formulae for this chapter. The last two appear in the formula booklet, but learn them too.

Sum of n ones: $\sum_{r=1}^{n} 1$

$= n.$

Sum of the first n integers: $\sum_{r=1}^{n} r$

$= \frac{1}{2}n(n+1).$

Sum of the first n squares: $\sum_{r=1}^{n} r^2$

$= \frac{1}{6}n(n+1)(2n+1).$

Sum of the first n cubes: $\sum_{r=1}^{n} r^3$

$= \frac{1}{4}n^2(n+1)^2.$

Notice:

$$\sum_{r=1}^{n} r^3 = \left(\sum_{r=1}^{n} r\right)^2.$$

The sum of the first n cubes equals the square of the sum of the first n integers.

Quickfire Triangulars!

Solution 5

In your head if you can...

$$\sum_{r=1}^{n} r = \frac{1}{2}n(n+1).$$

$1 + 2 + 3 + \cdots + 10$

$= 55.$

$11 + 12 + 13 + \cdots + 20$

$= 210 - 55 = 155.$

Sum up to 20, then remove everything up to 10:

$\left(\frac{1}{2} \times 20 \times 21\right)$

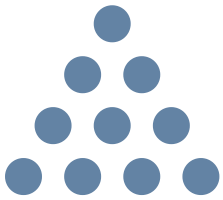
$- \left(\frac{1}{2} \times 10 \times 11\right).$

$1 + 2 + 3 + \cdots + 99$

$= 4950.$

$100 + 101 + 102 + \cdots + 200$

$= 20100 - 4950 = 15150.$



Further Examples of Sums of Natural Numbers

Solution 6

Use the formula $\sum_{r=1}^{n} r = \frac{1}{2}n(n+1)$ to evaluate these textbook examples.

$\sum_{r=1}^{4}(2r-1)$

$= 1 + 3 + 5 + 7 = 16.$

There are few enough terms to list them.

$\sum_{r=25}^{50} r$

$$= \sum_{r=1}^{50} r - \sum_{r=1}^{24} r$$

$= 1275 - \left(\frac{1}{2} \times 24 \times 25\right)$

$= 975.$

$\sum_{r=1}^{50} r$

$= \frac{1}{2} \times 50 \times 51 = 1275.$

Show $\sum_{r=5}^{2N-1} r = 2N^2 - N - 10$, for $N \geq 3$.

Tip: subtract the sum up to one less than the lower limit. Substitute the upper limit, here $2N-1$, for n.

$$\sum_{r=5}^{2N-1} r = \sum_{r=1}^{2N-1} r - \sum_{r=1}^{4} r$$

$= \frac{1}{2}(2N-1)(2N) - \frac{1}{2}(4)(5)$

$= 2N^2 - N - 10.$

Test Your Understanding So Far

Solution 7

Show that

$$\sum_{r=n}^{3n} r = 2n(2n+1).$$

Tip: after writing the subtraction, look for a common factor before expanding.

Proof

$$\sum_{r=n}^{3n} r = \tfrac{1}{2}(3n)(3n+1) - \tfrac{1}{2}(n-1)(n-1+1)$$
$$= \tfrac{3}{2}n(3n+1) - \tfrac{1}{2}n(n-1)$$
$$= \tfrac{1}{2}n(9n+3-n+1)$$
$$= \tfrac{1}{2}n(8n+4)$$
$$= 2n(2n+1).$$

Breaking Up Summations

Solution 8

Prove $\sum_{r=1}^{n} kr = k\sum_{r=1}^{n} r$, where k is constant.

Proof

$$\sum_{r=1}^{n} kr = k + 2k + \cdots + nk$$
$$= k(1 + 2 + \cdots + n)$$
$$= k\sum_{r=1}^{n} r.$$

Examples: $\sum_{r=1}^{n} 3r$

$$= 3\sum_{r=1}^{n} r = \tfrac{3}{2}n(n+1).$$

$\sum_{r=1}^{n} 4$

$$= 4n.$$

Breaking Up Summations

Solution 9

Prove $\sum_{i=1}^{n}(a_i + b_i) = \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i$.

Proof

$$\sum_{i=1}^{n}(a_i + b_i) = a_1 + b_1 + a_2 + b_2 + \cdots + a_n + b_n$$
$$= a_1 + a_2 + \cdots + a_n + b_1 + b_2 + \cdots + b_n$$
$$= \sum_{i=1}^{n} a_i + \sum_{i=1}^{n} b_i.$$

Combine this with the constant-multiple property:

$$\sum_{r=1}^{25}(3r + 1)$$

Example

$$= 3\sum_{r=1}^{25} r + \sum_{r=1}^{25} 1$$
$$= \tfrac{3}{2}(25)(26) + 25 = 1000.$$

Show $\sum_{r=1}^{n}(3r + 2) = \tfrac{n}{2}(3n + 7)$.

Further example

$$\sum_{r=1}^{n}(3r + 2) = 3\sum_{r=1}^{n} r + 2\sum_{r=1}^{n} 1$$
$$= \tfrac{3}{2}n(n + 1) + 2n$$
$$= \tfrac{n}{2}(3(n + 1) + 4)$$
$$= \tfrac{n}{2}(3n + 7).$$

Hence evaluate $\sum_{r=20}^{50}(3r + 2)$.

Hence

$$= \sum_{r=1}^{50}(3r + 2) - \sum_{r=1}^{19}(3r + 2)$$
$$= \tfrac{50}{2}(3(50) + 7) - \tfrac{19}{2}(3(19) + 7)$$
$$= 3317.$$

Exercise 3A — Core Pure Book 1 Pages 46–47

Solution 10

Exercise 3A — Core Pure Book 1, pages 46–47.

Exercise 3A — Core Pure Book 1, pages 46–47.

Exercise 3A — Core Pure Book 1, pages 46–47.

Exercise 3A — Core Pure Book 1, pages 46–47.

Exercise 3A — Core Pure Book 1, pages 46–47.

Sums of Squares and Cubes

Solution 15

(a) Show $\sum_{r=n+1}^{2n} r^2 = \frac{1}{6}n(2n+1)(7n+1)$.

(b) Verify for $n = 1$ and $n = 2$.

Tip: factor out common terms before expanding.

$$\sum_{r=1}^{n} r^2 = \tfrac{1}{6}n(n+1)(2n+1)$$

$$\sum_{r=1}^{n} r^3 = \tfrac{1}{4}n^2(n+1)^2.$$

The polynomial degree of each sum is one higher than that of its terms: squares give a cubic, cubes a quartic.

a

$$\sum_{r=n+1}^{2n} r^2 = \sum_{r=1}^{2n} r^2 - \sum_{r=1}^{n} r^2$$
$$= \tfrac{1}{6}(2n)(2n+1)(4n+1)$$
$$- \tfrac{1}{6}n(n+1)(2n+1)$$
$$= \tfrac{1}{6}n(2n+1)\big(2(4n+1) - (n+1)\big)$$
$$= \tfrac{1}{6}n(2n+1)(7n+1).$$

b

When $n = 1$:

$$2^2 = \tfrac{1}{6} \times 3 \times 8$$
$$4 = 4.$$

When $n = 2$:

$$3^2 + 4^2 = \tfrac{2}{6} \times 5 \times 15$$
$$25 = 25.$$

Test Your Understanding

Solution 16

Edexcel FP1 (Old), June 2013, Q5

(a) Use standard sums of r and r^2 to show

$$\sum_{r=1}^{n}(r+2)(r+3)$$
$$= \tfrac{1}{3}n(n^2 + 9n + 26)$$

for positive integers n. (6 marks)

(b) Hence show

$$\sum_{r=n+1}^{3n}(r+2)(r+3)$$
$$= \tfrac{2}{3}n(an^2 + bn + c),$$

finding integers a, b, c. (4 marks)

a

$$\sum_{r=1}^{n}(r+2)(r+3)$$
$$= \sum_{r=1}^{n} r^2 + 5\sum_{r=1}^{n} r + 6\sum_{r=1}^{n} 1$$
$$= \tfrac{1}{6}n(n+1)(2n+1)$$
$$+ \tfrac{5}{2}n(n+1) + 6n$$
$$= \tfrac{1}{6}n\big((n+1)(2n+1)$$
$$+ 15(n+1) + 36\big)$$
$$= \tfrac{1}{6}n(2n^2 + 18n + 52)$$
$$= \tfrac{1}{3}n(n^2 + 9n + 26).$$

b

$$\sum_{r=n+1}^{3n}(r+2)(r+3)$$
$$= \tfrac{1}{3}(3n)\big((3n)^2 + 9(3n) + 26\big)$$
$$- \tfrac{1}{3}n(n^2 + 9n + 26)$$
$$= n(9n^2 + 27n + 26)$$
$$- \tfrac{1}{3}n(n^2 + 9n + 26)$$
$$= \tfrac{1}{3}n(27n^2 + 81n + 78$$
$$- n^2 - 9n - 26)$$
$$= \tfrac{1}{3}n(26n^2 + 72n + 52)$$
$$= \tfrac{2}{3}n(13n^2 + 36n + 26).$$

$$a = 13,\ b = 36,\ c = 26.$$

Harder Exam Question

Solution 17

Edexcel FP1 (Old), January 2011, Q5

(a) Use the sums of r, r^2, r^3 to prove

$$\sum_{r=1}^{n} r(r+1)(r+5) = \tfrac{1}{4}n(n+1)(n+2)(n+7)$$

for all positive integers n. (5 marks)

(b) Hence, or otherwise, find $\sum_{r=20}^{50} r(r+1)(r+5)$. (2 marks)

a

$$\sum_{r=1}^{n}(r^3+6r^2+5r)$$
$$= \tfrac{1}{4}n^2(n+1)^2 + 6\times\tfrac{1}{6}n(n+1)(2n+1)$$
$$+ 5\times\tfrac{1}{2}n(n+1)$$
$$= \tfrac{1}{4}n^2(n+1)^2 + n(n+1)(2n+1) + \tfrac{5}{2}n(n+1)$$
$$= \tfrac{1}{4}n(n+1)\big(n(n+1)+4(2n+1)+10\big)$$
$$= \tfrac{1}{4}n(n+1)(n^2+n+8n+4+10)$$
$$= \tfrac{1}{4}n(n+1)(n^2+9n+14)$$
$$= \tfrac{1}{4}n(n+1)(n+2)(n+7).$$

b

Let $S_m = \sum_{r=1}^{m} r(r+1)(r+5)$. Then

$$\sum_{r=20}^{50} r(r+1)(r+5) = S_{50} - S_{19}$$
$$= \tfrac{1}{4}(50)(51)(52)(57)$$
$$- \tfrac{1}{4}(19)(20)(21)(26)$$
$$= 1889550 - 51870$$
$$= 1837680.$$

Exercise 3B — Core Pure Book 1 Pages 49–50

Solution 18

Exercise 3B — Core Pure Book 1, pages 49–50.

Exercise 3B — Core Pure Book 1 Pages 49–50 Solution 19

Exercise 3B — Core Pure Book 1, pages 49–50.

Exercise 3B — Core Pure Book 1 Pages 49–50 Solution 20

Exercise 3B — Core Pure Book 1, pages 49–50.

Exercise 3B — Core Pure Book 1, pages 49–50.

Exercise 3B — Core Pure Book 1, pages 49–50.

Bonus conundrum

Given that n is even, determine

$1^2 - 2^2 + 3^2 - 4^2 + 5^2 - \cdots - n^2$.

Sum

$$= \sum_{r=1}^{n/2} (2r-1)^2 - \sum_{r=1}^{n/2} (2r)^2$$

$$= \sum_{r=1}^{n/2} (1 - 4r)$$

$$= \tfrac{n}{2} - 4 \cdot \tfrac{1}{2} \cdot \tfrac{n}{2} \left(\tfrac{n}{2} + 1\right)$$

$$= -\tfrac{1}{2} n(n+1).$$

Alternative

Alternatively, each pair is a difference of squares:

$$(3 \times -1) + (7 \times -1) + (11 \times -1) + \cdots + ((2n-1) \times -1)$$

$$= -1\big(3 + 7 + 11 + \cdots + (2n-1)\big).$$

The brackets contain an arithmetic series with first term 3, difference 4 and $n/2$ terms:

$$-\frac{n/2}{2}\big(3 + (2n-1)\big)$$

$$= -\tfrac{1}{2} n(n+1).$$